

Ahmed Ikram | Data Engineering & Analytics

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PROFILE

Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree, graduating June 2027. Designs and operates data infrastructure end-to-end - streaming ingestion, warehouse/lakehouse modeling, and the orchestration and observability layers that keep pipelines trustworthy - across three genuinely different paradigms: broker-mediated event streaming on AWS, batch lakehouse ETL on Azure, and broker-less real-time OLAP streaming on GCP. Recent work: a self-hosted ClickHouse platform ingesting 58.9M+ real-time Wikipedia edit events at up to 5,655 events/sec with zero data loss across 8 chaos-engineering injections, and a Databricks medallion pipeline processing 153K+ IIS log rows through 7 automated quality gates into a Kimball-style star schema. Comfortable being the data infrastructure a machine learning system depends on, not just the pipeline behind a dashboard. Seeking a graduate **Data Engineering** role building systems that stay correct and observable under real load.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree. Expected Graduation: **June 2027**

High School of Dundee

Aug 2017 - June 2023

- Advanced Higher Computer Science (C), Higher Maths and English (A), Higher Chemistry (B) and Higher Physics (C)

TECHNICAL SKILLS

Languages: Python · SQL · TypeScript · JavaScript · Bash

Pipeline Orchestration & Transformation: Apache Airflow · Apache Kafka · dbt · Apache Spark · PySpark · Pydantic (Data Contracts)

Streaming & Real-Time Systems: ClickHouse · Materialized Views (SummingMergeTree) · Server-Sent Events · Idempotent Consumers · At-Least-Once Delivery

Data Modeling & Warehousing: Kimball Dimensional Modelling (Star Schema) · Slowly Changing Dimensions (Type 1) · Delta Lake · BigQuery · Databricks (DLT, Unity Catalog)

Data Quality & Observability: Great Expectations · dbt Tests · Grafana · Prometheus · Azure Monitor · Cloud Monitoring · StatsD

Databases: PostgreSQL · MongoDB · MySQL · SQLite · Redis · Alembic

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, IAM, CloudFront, SageMaker) · Azure (Databricks, Azure SQL, Data Factory, ADLS) · Google Cloud (Compute Engine, BigQuery, GCS, Secret Manager, Workload Identity Federation) · Terraform · Docker · GitHub Actions

Testing: Pytest · dbt Tests · Great Expectations · Terraform (Checkov)

PROJECTS

W3C Web Logs ETL Pipeline | Apache Airflow · Python · Databricks · dbt · Apache Spark · PySpark · Azure SQL · Delta Lake · Terraform · Docker · Power BI · DAX · Power Automate · Grafana · Prometheus

- Ingested 93 W3C IIS log files (155,380 rows, 0 dropped) into a Databricks DLT medallion pipeline (Bronze → Silver) on via Auto Loader, with a custom parser detecting 14- vs 18-field IIS formats at parse time and 7 DLT quality gates; Silver enriches every row via a consolidated MaxMind GeoIP struct UDF (3.5× faster than 7 separate UDFs), achieving 99.99% known-country coverage across 153,377 enriched, 31-column rows
- Modeled a **Kimball-style star schema** across 9 dimension tables: geolocation and user-agent dimensions apply **SCD Type 1** (SQL MERGE upsert on SHA-256-hashed natural keys) with -1 sentinel rows preserving FK integrity on unmatched lookups; 16 dbt models (10 staging + 6 marts) compile against both PostgreSQL and T-SQL from one codebase via 18 compatibility macros, with an incremental fact table deduping on a 12-component MD5 key that caught 213 duplicate rows in the raw feed
- 4 Airflow DAGs orchestrate the full pipeline with zero processing logic in DAG files - layers decoupled via Airflow Datasets, no ExternalTaskSensor; 30+ Azure resources provisioned across 2 Terraform configurations including the full

OIDC auth chain (Azure AD app, service principal, federated credential, role assignment) via a single terraform apply, with a rollback workflow that re-applies Terraform from the previous commit on demand

- 121 dbt data tests (not_null, unique, accepted_values, relationships, expression_is_true) gate every model before Power BI; a custom Python freshness probe feeds 3 Grafana dashboards (23 panels) and 10 alert rules across 3 severity tiers routed to Slack; 613 tests across 5 tiers, including dual-dialect dbt CI against side-by-side PostgreSQL and SQL Server containers, back a 7-page Power BI dashboard (155K requests, 88 countries, P95 1,149ms)

ATM Log Aggregation, Analysis & Diagnostics Platform | *FastAPI, Apache Kafka, PostgreSQL, Redis, MLflow, SageMaker, Docker, Terraform, AWS*

- Data engineering lead on a 7-person Agile team for NCR Atleos - a real industry stakeholder, not a solo project; independently replaced a direct-DB generator with an event-driven Apache Kafka KRaft ingestion pipeline processing 2.5M+ events every day at 100+ msg/sec across 7 simultaneous log sources into 2 topics (3 partitions each); 7 source-specific parsers with safe-default field access ensure schema drift never halts ingestion and malformed records route to a dead-letter table
- Consumer delivery is **at-least-once with application-level idempotent deduplication** - hybrid Redis SET+TTL primary and 10K-entry LRU fallback - holding effectively-once behaviour across consumer restarts; failed messages retry via a Redis Stream DLQ with exponential backoff (5s → 10s → 20s, max 3 retries) before landing in a dead-letter table for manual triage
- PostgreSQL data lake on JSONB payloads (new log sources need only a new parser, zero schema migrations), with a `v_unified_analysis UNION ALL` view and COALESCE field normalisation as the single query target for both the ML detection engine and the ChromaDB-backed RAG assistant; 14 composite B-tree indexes on (timestamp, entity, type) sustain sub-100ms queries past 2M+ rows, served by a ThreadedConnectionPool (min=5, max=50) with 3-retry backoff across concurrent API, detection, and cleanup threads
- MLflow (RDS backend + S3 artifacts) tracks every training run with champion-alias promotion and auto-retrain on startup if artifacts are missing or version-skewed; full AWS estate (10 Terraform modules, 118 resources: multi-AZ VPC, ECS Fargate, RDS, S3, CloudFront, 6 least-privilege IAM roles) validated by 1,402 tests catching silent data loss under 50 concurrent writes

WikiStream Real-Time Streaming Analytics Platform | *Python · ClickHouse · Grafana · BigQuery · Terraform · Google Cloud · Pydantic · Great Expectations · sysetmd · Github Actions*

- Async Python consumer streams Wikipedia's public EventStreams SSE feed (hand-rolled WHATWG-compliant parser, zero-dependency) into a self-hosted **ClickHouse 26.3** cluster - **58.9M+ events ingested**, sustaining 203.5 events/sec average with real-world peaks of 2,719 events/sec; Pydantic v2 schema-on-write validation rejects malformed events to a dead-letter table before they ever reach disk
- At-least-once delivery via a durable, atomically-written SSE cursor (Last-Event-ID resume) and a dedup ring buffer absorbing upstream redeliveries - proven with a live SIGKILL-mid-insert test that resumed with zero loss and zero duplicates; an 8-injection chaos battery (consumer kill, DLQ flood, ClickHouse outage, disk-fill to 86%, firewall lockdown) fired and cleared every one of 5 Grafana + 2 Cloud Monitoring alert rules
- 3 SummingMergeTree materialized views pre-aggregate edits-per-minute, top-pages, and edit-size distributions at ingestion time, cutting dashboard query latency **15.0x at p50** (13.2x p99) and rows scanned ~200x (46.8M → 0.23M) versus querying raw events directly - verified through a burst-test harness that sustained 5,655 events/sec for 60s (2.08x real observed peak) across 577,738 events with zero drops
- Hourly ClickHouse → GCS → BigQuery export keeps a 5-table partitioned warehouse tier queryable past the 30-day raw TTL, with parity verified via **SUMS comparison - not row counts**, which are merge-state-unsafe on SummingMergeTree - confirming exact matches (e.g. 48,833 edits, 43,943,656 bytes) on every hourly run; a Great Expectations suite (11 checks: freshness lag, null domains, cardinality, bot-ratio bounds) runs hourly on a 5% sample and gates a pipeline_health verdict
- Terraform-provisioned on GCP with Workload Identity Federation (zero static credentials) and a 22-row least-privilege IAM review matrix; itemized FinOps tracking (not an estimate) held the full pipeline to **\$41.65/month** run-rate - \$1.79/month residual post-teardown - validated by 143 tests at 99.38% coverage (100% on 6 business-critical modules) with a CI gate proven to actually block regressions

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations

Feb 2025

AWS Academy Graduate - Cloud Foundations

Mar 2024

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

- Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023

- Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure